Performance Evaluation of Machine Learning Models for Intrusion Detection in MQTT-based IoT Environments

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

What is the most effective machine learning approach for intrusion detection in privacy-sensitive, resource-constrained IoT environments using the MQTT protocol? In this study, we compare and contrast many machine learning models that have been developed for use in intrusion detection systems (IDS) in IoT settings that rely on the Machine Data Transfer Protocol (MQTT). The study utilizes the real-world, balanced **MQTTEEB-D 2025** dataset to thoroughly evaluate tree-ensemble classifiers (**XGBoost**), neural networks (**MLPs**, **GRUs**), and transformer-based models (**TabTransformer**) in both centralized and federated learning environments.
We evaluate privacy-preserving federated techniques, including a novel reliability-weighted aggregation method (**FLAIR-XGB**), to assess their efficacy with heterogeneous data and in scenarios typical of IoT installations. The concept of knowledge distillation from ensemble teachers to lightweight neural students is analyzed to enable efficient edge deployment.
Findings demonstrate that centralized and federated **XGBoost** models exhibit superior accuracy (about 88–89%), while distilled **MLPs** present a favorable balance between performance and resource constraints. The research highlights the practical benefits of federated learning in terms of privacy, scalability, and real-time detection within IoT healthcare and industrial settings.

Keywords: IoT security, intrusion detection system, MQTT, federated learning, XGBoost, machine learning evaluation, knowledge distillation, edge deployment, privacy preservation, IoMT cybersecurity

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Chapter 1

Introduction

1.1 Background

The recent developments in the Internet of Things (IoT) have had an effect on the network infrastructure, where billions of smart devices can now be connected in different sectors, like in healthcare, manufacturing, and urban development. The Message Queuing Telemetry Transport (MQTT) protocol has found popular use in an IoT implementation due to its lightweight, publish-subscribe protocol, allowing fast and energy-efficient communication between resource-constrained devices (Banks et al., 2019). However, the risk of advanced cyberattacks on MQTT-enabled systems has increased due to the higher network connectivity and system diversity, so it has become a mandatory condition that effective intrusion detection is a necessity (Almiani et al., 2020). The distributed and decentralized characteristics of IoT/MQTT models provide a security challenge, particularly in the limited resources with privacy as the main parameter. The machine learning-based IDS solutions were proved to be helpful in terms of the adaptive monitoring and anomaly detection, but still there exist a lot of issues, including the secure communication of sensitive information, the computational limitations at the edge, and the possibility of robust performance in the case of various attack patterns (Nguyen et al., 2022). Despite the proven potential of Machine Learning (ML)-based Intrusion Detection Systems (IDS), achieving robust and reliable performance for security in MQTT-based IoT environments remains challenging. A major hindering factor is the historical lack of a representative, balanced, real-world dataset that accurately captures the specific attack patterns and genuine threats associated with the MQTT protocol. This scarcity of relevant "ground truth" data makes it difficult to rigorously benchmark and validate ML models that can be trusted to handle evolving cyber-threats in practical IoT deployments. Here comes the importance of MQTT-specific datasets, which enable accurate training, evaluation, and benchmarking of IDS models tailored for real-world IoT threat scenarios.
Figure **??** illustrates the core security challenges in distributed MQTT-based IoT environments and how machine learning-based Intrusion Detection Systems (ML-based IDS) are employed for adaptive monitoring while facing obstacles related to data privacy, computational limits, and detection robustness.

1.2 Rational of the Study or Motivation

The service of communication in interconnected IoT networks can be described as a requirement element of society and industry. The most common type of legacy IDS solution needs thorough access to sensitive data, which can be inconsistent with privacy laws like GDPR and HIPAA (Gubbi et al., 2013). Numerous state-of-the-art machine learning systems are not computationally efficient enough to be directly implemented on commonly available IoT hardware (Sun et al., 2022). Exposed to certain vulnerabilities, the MQTT protocol has weaknesses in its session management and unauthorized publish/subscribe functions, which generic solutions do not sufficiently deal with (Alrashdi et al., 2019). These gaps require custom IDS models that can work under limited conditions and at the same time meet the privacy and real-time operational requirements.

1.3 Problem Statement

Although machine learning-based IDS have been proven to be effective at general anomaly detection, they cannot be applied to federated and privacy-preserving environments due to a lack of benchmarking and cannot be applied to real-world MQTT threats due to the lack of standard datasets (Fahmideh et al., 2018). The big question that will be addressed in this thesis is how to successfully analyze IDS models of MQTT-based IoT that will compromise on accuracy, efficiency, privacy, and feasibility to deploy IoT.

1.4 Objective

The objectives of this research are summarized as follows:

- Compare the XGBoost, MLP, GRU, and TabTransformer performance on the MQTTEEB-D 2025 dataset (Wu et al., 2020).
- Compare and contrast centralized and federated methods of learning, such as the FLAIR-XGB aggregation method, considering their capacity to preserve privacy in the heterogeneous setting (Kairouz et al., 2019).
- Understand knowledge distillation methods, which can be used to construct high-performance but lightweight models that can be deployed to edge devices (Gou et al., 2021).
- Carry out special MLP experiments in the context of MQTT network security.
- Explore privacy, scalability, and real-time intrusion detection trade-offs in a healthcare and industrial IoT-based setting.
- Systematically analyze model robustness under operational stress conditions and evaluate resource efficiency to ensure practical deployment in real-world IoT environments.

1.5 Methodology in Brief

The experiment design involves the acquisition and preprocessing of the MQTTEEB-D 2025 dataset, the training of the baseline models in the context of centralized and federated models, and the evaluation of the results through conventional performance metrics (Abes et al., 2025). The focus is on federated learning variants and knowledge distillation, and all the models are tested in the context of the simulated heterogeneity and drift of the data. Comparative evaluation is based on statistical analysis and ablation studies (Li et al., 2020). Performance was evaluated using macro-averaged F1 scores, per-class precision and recall, computational efficiency metrics, and robustness tests simulating client drift, label noise, and dropout in federated settings.

1.6 Scopes and Challenges

The given study specifically dwells on MQTT-powered IoT and pays attention to privacy and resource limits in the context of healthcare and industry (Almiani et al., 2020). It is not applicable to other IoT protocols. Such challenges are federated edge hardware, handling data heterogeneity, and trade-offs between model complexity and resource and privacy demands (Kairouz et al., 2019).

1.7 Team Overview

Project work was divided into tasks carried out by team members with expertise in literature review, methodology, implementation, and evaluation. Regular meetings and collaboration ensured high standards and the constant improvement of the methodology.

1.8 Key Learnings and Insights

The work also increased the team's work on MQTT security, federated learning, and privacy-preserving analytics. Lessons learned include trade-offs of aggregation strategy, governance of data, advanced experimental design, and successful collaboration.

Chapter 2

Literature Review

2.1 Literature Review and Background

2.2 Preliminaries

Intrusion detection in IoT environments utilizing the MQTT protocol is a multi-disciplinary challenge, drawing from networking, cybersecurity, and advanced machine learning. The MQTT protocol, with its lightweight publish-subscribe model, is widely used for efficient communication among resource-constrained devices, but this efficiency introduces vulnerabilities that can be exploited by sophisticated cyber threats [2].
Machine learning-based intrusion detection systems (IDS) have emerged as promising solutions, leveraging tree ensembles, deep neural networks, and transformer-based architectures to identify complex attack patterns [9]. The evolution of privacy-preserving and federated learning approaches has further shifted the focus toward protecting sensitive device data while maintaining high detection accuracy [3].
Key foundational concepts include:

- **MQTT Protocol and IoT Architecture:** Overview of MQTT operations, IoT topologies, and protocol-specific attack vectors [4].
- **Supervised and Ensemble ML:** Principles of XGBoost, MLP, GRU, and TabTransformer, and their IDS applications [6].
- **Federated Learning:** Distributed model training with privacy, using parameter aggregation instead of raw data [5].
- **Knowledge Distillation:** Training efficient student models from teacher models, enabling lightweight edge deployment [8].

2.3 Review of Existing Research

Recent research has accelerated the development of ML-powered IDS for MQTT-enabled IoT, driven by technological advances and increasing IoT-specific cyber threats. Early works highlighted shortcomings of rule- or signature-based IDS for dynamic IoT traffic [1]. Later studies emphasized MQTT-specific vulnerabilities (e.g., insecure topic management, DoS floods, authentication bypass), motivating

custom lightweight IDS frameworks [4]. Balanced datasets such as **MQTTEEB-D 2025** have become critical for reproducible and trustworthy benchmarking [16].

Model-Specific Advances

XGBoost (Tree Ensemble): XGBoost is a leading approach for tabular IDS, balancing interpretability and accuracy. Studies report 88–89% accuracy on IoT datasets, though deployment on constrained devices remains difficult due to resource usage [6], [10].
Neural Models (MLP, GRU): MLPs are computationally efficient and useful in distilled variants, while GRUs capture temporal dependencies in traffic and attack sequences [9], [10]. With proper tuning, they achieve competitive accuracy with lower inference costs.
TabTransformer and Deep Tabular Models: Transformer-based architectures like TabTransformer leverage contextual embeddings to model categorical-numerical features. These models can outperform ensembles when sufficient data diversity exists [6], [11], [13].
Hybrid and Federated Models: Hybrid CNN-LSTM/GRU frameworks, often with attention, represent deep-learning state-of-the-art for IoT anomaly detection [14]. Federated learning methods (FedAvg, FLAIR-XGB) extend IDS training into privacy-sensitive environments, improving resilience to heterogeneity without sharing raw data [3], [12].
Knowledge Distillation in Federated Learning: Recent works (e.g., FedKD-IDS, FD-IDS) merge KD with FL to transfer soft outputs from ensemble teachers into lightweight students, reducing communication cost and complexity while maintaining near-teacher accuracy [8], [15], [17].

Dataset Developments

The MQTTEEB-D 2025 dataset enables realistic, balanced benchmarking of MQTT IDS. It covers diverse attack types (SlowITe, Flood, Brute Force, Injection) in balanced form, supporting both centralized and federated research under heterogeneous IoT settings [16].

2.4 Summary of Key Findings

The literature shows that no single ML model is universally optimal for MQTT IDS. Tree ensembles (XGBoost) consistently deliver strong accuracy in both centralized and federated setups [6], [10], [11], [13], [16]. Transformer-based and hybrid models can exceed ensembles on complex data but require more compute [7], [14]. For deployment, distilled student models strike a balance between accuracy and efficiency [8], [15], [17]. Federated aggregation methods (FedAvg, FLAIR-XGB) enhance privacy and scalability, retaining robustness under data drift [3], [12]. Real-world balanced datasets are indispensable for rigorous evaluation and future progress.

Chapter 3

Requirements, Impacts and Constraints

3.1 Specifications, Impact, and Resource Management

The project will reduce the development of a powerful MQTT-based IoT environment IDS with clear objectives, technical specifications, and adherence to applicable standards. The first one aims to develop and test machine learning models that can be used to deliver effective and precise intrusion detection in privacy-sensitive limited resource networks.
Functional requirements include real-time detection of anomalies, compliance with the MQTT protocol, decentralization or federation, and heterogeneous data flexibility for edge devices. Non-functional requirements are focused on safeguarding privacy, low cost of calculations, scalability, and resistance to concept drift (Li et al., 2020; Kairouz et al., 2019). The project is based on the requirements of GDPR, ISO/IEC 27001, and MQTT. Robustness to client drift, label noise, and dropout, as well as efficiency in terms of training time, inference latency, and communication overhead, are also considered essential non-functional requirements for the proposed IDS.
Figure 3.1 provides an overview of the final specifications and requirements for the proposed IDS system.

3.2 Societal Impact

The IoT environments that use MQTT can be extremely significant to society in the context of their security. An effective IDS guarantees patient privacy and safety in the healthcare environment, ensures operational continuity, and prevents disruptions in industry (Bharot and Breslin, 2023; Almiani et al., 2020; Gubbi et al., 2013).
The project assists with building digital trust and compliance with regulations, so advanced intrusion detection can be made available to organizations of any size, including those with limited resources (Sun et al., 2022). Secure IoT networks allow development of new technologies and facilitate the adoption of smart technologies in various fields (Nguyen et al., 2022).
Figure 5.6 illustrates the societal implications of secure MQTT-enabled IoT systems.

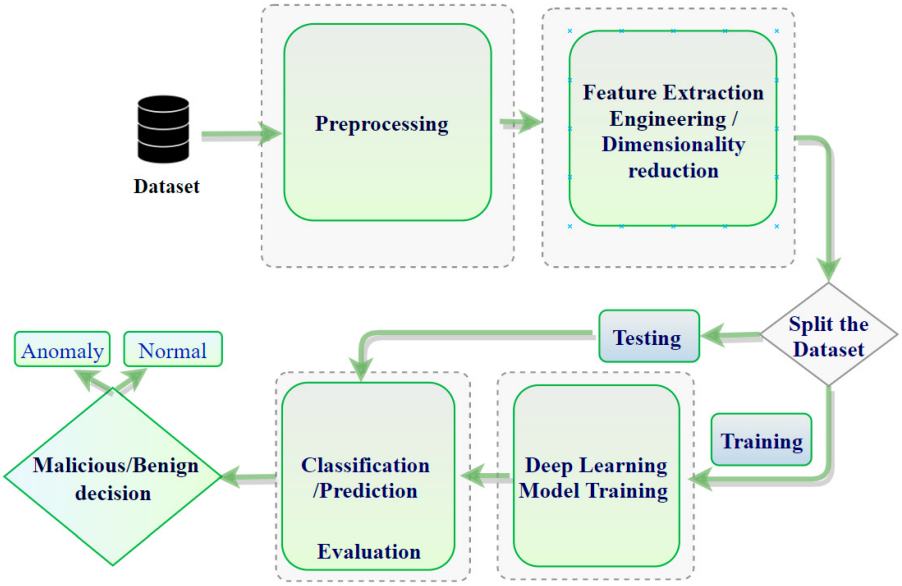


Figure 3.1: Overview of final specifications and requirements for the proposed MQTT-based IDS system.

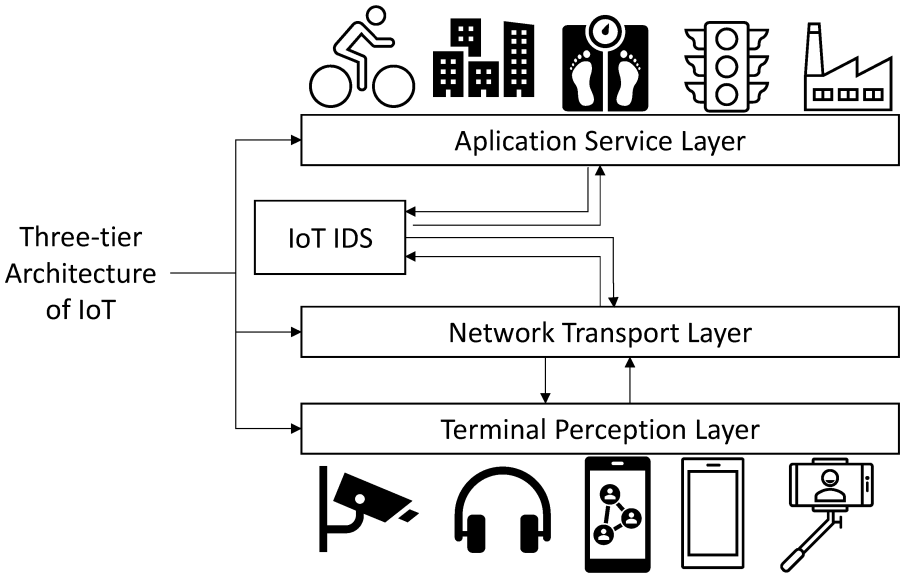


Figure 3.2: Societal impact of secure MQTT-enabled IoT systems.

3.3 Environmental Impact

IoT security solutions are closely linked to sustainability. Lightweight machine learning models use less energy and generate less electronic waste because they consider the lifecycle of edge devices (Gubbi et al., 2013; Sun et al., 2022).
Federated learning further improves sustainability by distributing computation, reducing server overload, and cutting overall power expenditure (Kairouz et al., 2019). The project also measures resource savings compared to traditional centralized IDS, demonstrating its contribution to environmentally friendly digital infrastructure.
Figure 5.7 highlights the environmental impact and sustainability advantages of the proposed IDS.

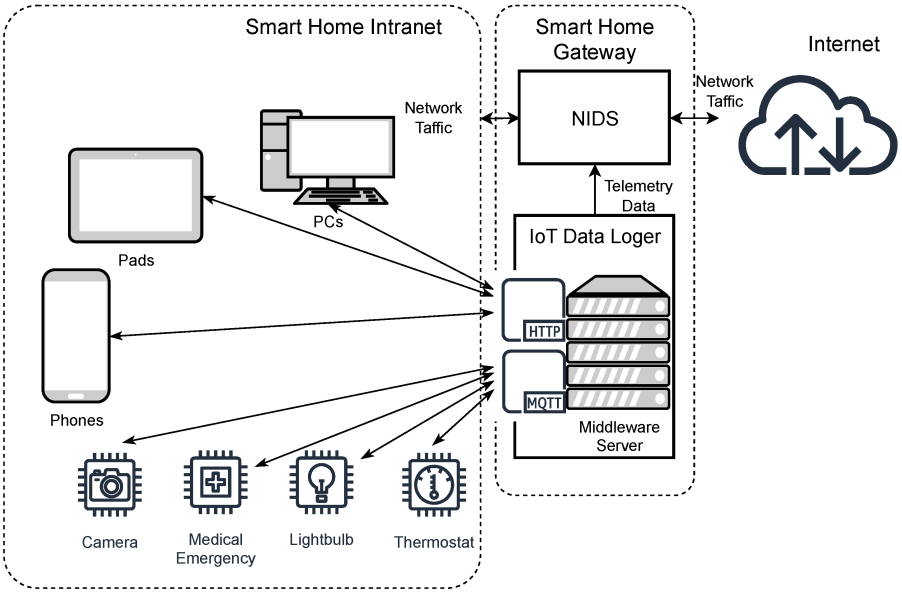


Figure 3.3: Environmental impact and sustainability benefits of lightweight federated IDS models.

3.4 Ethical Issues

The design and implementation of IoT-based IDS involves multiple ethical considerations. The project complies with the principles of academic integrity, applies peer-reviewed literature exclusively, and implements reproducible methodologies. Privacy is prioritized through data anonymization and federated learning, ensuring no direct data sharing is performed (Kairouz et al., 2019; Gou et al., 2021).
The choice of MQTTEEB-D 2025 is ethically justified as it is balanced and non-discriminatory (Abes et al., 2025). All models are tested for fairness, and deployment strategies are verified to prevent bias or misuse. The team also responds proactively to adversarial risks, ensuring robust security and verifiability.
Figure 3.4 summarizes the ethical principles guiding the design and deployment.

3.5 Standards - if applicable

The project aligns with GDPR regulations, ISO/IEC 27001 standards, and MQTT protocol compliance, ensuring industry alignment and legal adherence.

3.6 Project Management Plan

The project management plan follows progressive stages:

1. Needs assessment and literature research.
2. Data processing and acquisition.
3. Model selection and prototyping (XGBoost, MLP, GRU, TabTransformer).
4. Implementation of federated learning and distillation protocols.

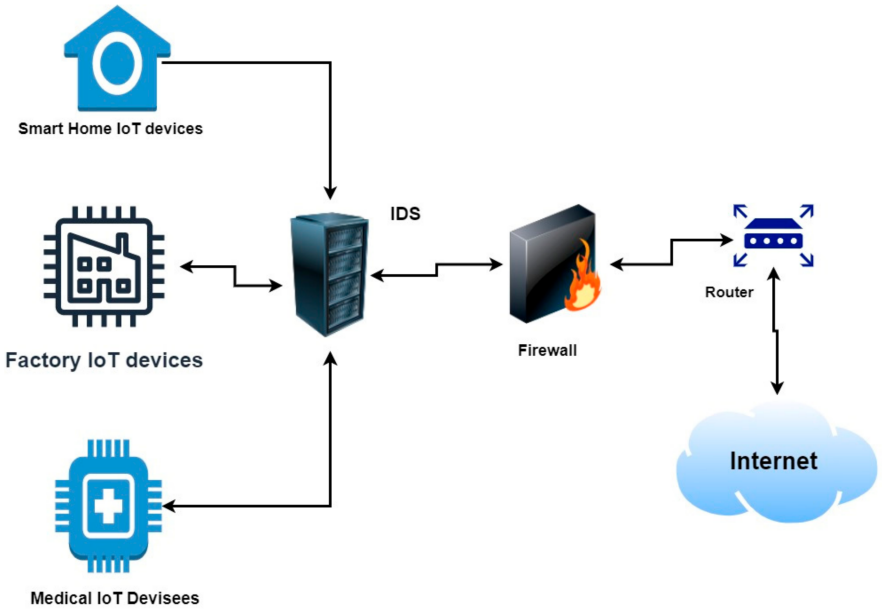


Figure 3.4: Ethical considerations in the design and deployment of the IDS.

5. Rigorous assessment and benchmarking.

6. Compilation of reports and stakeholder review.

Resource distribution is based on task complexity, with milestones and schedules set at the start. Regular meetings are held for progress tracking and reporting. Open-source tools and remote collaboration help manage budget constraints. All experimental conditions, hyperparameter settings, and code repositories are documented to ensure reproducibility and transparency.

3.7 Risk Management

The project incorporates a strong risk management strategy. Key risks include data privacy breaches, resource constraints, concept drift, and project delays. Mitigation strategies include enhanced anonymization, resource optimization, frequent hyperparameter tuning, balanced data usage, and contingency planning. Continuous supervisor feedback is also used to navigate risks effectively.

3.8 Economic Analysis

The economic analysis considers costs of hardware, software, development, and maintenance. By leveraging lightweight, federated, and distilled models, costs are minimized through reduced server loads, bandwidth, and hardware upgrades (Sun et al., 2022; Okey et al., 2023). Open-source software and cloud applications further lower startup costs, making the solution accessible to small-scale implementations.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

This chapter presents the comprehensive research methodology adopted to investigate intrusion detection within MQTT-based IoT environments using machine-learning techniques. The methodological framework was designed to achieve three complementary goals: (1) empirical performance benchmarking of different model families, (2) evaluation of privacy-preserving federated learning strategies, and (3) assessment of robustness and deployment feasibility under realistic IoT constraints. The overall design integrates centralized and federated learning paradigms to balance privacy preservation with detection efficacy. A systematic experimental pipeline was followed, covering dataset selection, preprocessing verification, model configuration, training, evaluation, and reproducibility validation. Each stage was implemented with statistical rigor to ensure that the observed differences in performance stem solely from the learning paradigms rather than from data preprocessing biases.

4.2 Preliminary Design or Model Specification

The design phase involved selecting and configuring multiple candidate architectures suited for MQTT-based intrusion detection. Three principal model families were included:

- **Tree-Ensemble Models (XGBoost):** Chosen for interpretability and high predictive performance on tabular data. Serves as baseline and teacher in the knowledge-distillation framework.
- **Neural Architectures (MLP, GRU):** Captures non-linear and sequential behavior in MQTT traffic. The GRU models temporal dependencies; MLP acts as a lightweight student model.
- **Transformer-Based Models (TabTransformer V2):** Leverages self-attention on mixed categorical–numerical MQTT features.

Both centralized and federated configurations were explored. A custom **FLAIR-XGB aggregation** was introduced to address client heterogeneity, drift, and unreliable participation. This extends FedXGB by weighting client contributions using local macro-F1, entropy, class balance, and drift stability.

4.3 Data Collection and Preparation

4.3.1 Sources of Data and Methods of Collection

The study employs the **MQTTEEB-D 2025 dataset** [16], a real-world benchmark for MQTT-protocol traffic analysis. Collected from interconnected medical and industrial IoT testbeds communicating via MQTT brokers, it is publicly available on Mendeley Data.[1]

Dataset Structure and Selection Rationale

The dataset repository provides multiple processed variants: `All_Processed.csv`, `Processed_normalized.csv`, `Processed_standardized.csv`, and `Processed_SMOTE.csv`. This research used the **SMOTE-balanced version** for three reasons:

1. **Fair Benchmarking:** Official balanced benchmark ensures comparability with prior work.
2. **Federated Robustness:** Prevents minority-class scarcity during client partitioning.
3. **Focus on Paradigms:** Removes resampling artifacts, isolating model differences.

Dataset Characteristics

Table 4.1 summarizes the dataset's class distribution and feature types.

Table 4.1: MQTTEEB-D 2025 Dataset Statistics – Balanced Class Distribution

Class Name	Samples	Percentage	Feature Type
Brute Force	105,774	16.67%	MQTT Flags
DoS	105,774	16.67%	Network Traffic
Flood	105,774	16.67%	Flow Features
Legitimate	105,774	16.67%	Normal MQTT Flow
Malformed Packets	105,774	16.67%	Protocol Violation
SlowITe	105,774	16.67%	Timing-Based Attack

Each record includes 11 MQTT-specific attributes (`tcpflags`, `mqttkalive`, `mqttconflags`, `mqttqos`, etc.).

4.3.2 Data Cleaning

As the SMOTE-balanced dataset was preprocessed by its authors, no additional cleaning was required. Metadata checks confirmed encoding consistency and completeness.

[1] https://data.mendeley.com/datasets/jfttfjn6tr

4.3.3 Data Transformation

Continuous features were normalized to [0,1]; categorical flags were label-encoded. All transformations followed the dataset's validated pipeline.

4.3.4 Data Integration

For federated simulations, the dataset was split into 3 clients via StratifiedKFold, maintaining equal class representation.

4.3.5 Data Reduction

All 11 features contributed materially. Exploratory PCA (95% variance) confirmed redundancy-free design.

4.3.6 Summary of Preprocessed Data

Final dataset: 634,644 samples, 6 balanced classes, 11 features. Data split: 80:20 train–test, with 10% validation in training. Federated partitions applied identical splits.

4.4 Implementation of Selected Design

4.4.1 Hardware and Software Environment

Table 4.2 shows the experimental environment.

Table 4.2: Experimental Environment Specifications

Component	Specification	Version / Model
GPU	NVIDIA RTX 2080 Ti	11 GB VRAM (CUDA 12.2)
CPU	AMD Ryzen 9 3900X	12 Cores / 24 Threads @ 3.8GHz
RAM	DDR4 3200 MHz	80 GB
OS	Ubuntu 22.04 LTS	–
XGBoost	Gradient Boosting Library	v1.7.6
TensorFlow	Deep Learning Framework	v2.12.0
Python	Programming Language	v3.10

4.4.2 Model Configurations

(a) Centralized XGBoost – Teacher Model Objective: `multi:softprob` (6 classes), 200 rounds, depth=8, lr=0.05. Baseline performance and KD teacher.

(b) Federated XGBoost (FedXGB) Local XGBoost per client with identical hyperparameters. Aggregated via FedAvg (size-weighted average).

(c) FLAIR-XGB (Proposed) Reliability-weighted aggregation:

$$R_i = \alpha F1_i + \beta(1 - H_i) + \gamma B_i - \delta D_i$$

Where $F1_i$=macro-F1, H_i=entropy, B_i=class balance, D_i=drift penalty. Only inter-quartile reliable clients contribute.

(d) Knowledge-Distilled MLP (Student) Teacher = XGBoost. Student = lightweight MLP. Loss function:

$$L = (1 - \lambda)L_{CE} + \lambda T^2 L_{KD}$$

(e) GRU and TabTransformer Variants GRU: 64 hidden units, dropout=0.3. TabTransformer V2: 6 layers, 8 heads, 128-dim embeddings.

4.5 Workflow and Validation Framework

Steps: 1. Dataset verification and partitioning. 2. Model initialization with documented hyperparameters. 3. Training with early stopping, k=5 CV. 4. Federated aggregation/sync. 5. Evaluation: accuracy, precision, recall, F1, efficiency. 6. Explainability: SHAP feature importance.
Random seeds, dataset hashes, and model checkpoints logged for reproducibility. Each experiment repeated 3 times; mean results reported in Chapter 5.
Summary: This chapter established a reproducible framework integrating multiple model families, learning paradigms, and aggregation strategies. By fixing preprocessing to the SMOTE dataset, the analysis focused solely on architectural and federated-learning choices for IDS in MQTT IoT environments.

Chapter 5

Result Analysis

5.1 Performance Evaluation

This chapter presents a comprehensive experimental evaluation of machine learning methods for federated intrusion detection in MQTT-based IoT environments. The assessment systematically compares centralized XGBoost, federated XGBoost (FedXGB), the proposed FLAIR-XGB aggregation method, knowledge-distilled neural networks (MLP), TabTransformer variations, and GRU models. The evaluation considers not only classification accuracy, but also efficiency, robustness, and interpretability, leveraging the MQTTEEB-D 2025 dataset to ensure realism and balance in empirical testing.

5.1.1 Evaluation Criteria

Performance was assessed across a spectrum of empirically relevant dimensions:

- **Classification Accuracy and Per-Class Recall:** Primary measure of model effectiveness for intrusion detection across multiple attack and normal categories.
- **Macro-Averaged F1 Score and Per-Class Precision:** Balanced assessment under class imbalance, with focus on less-represented attack types.
- **Training and Inference Efficiency:** Total duration and memory usage measured for computational practicality.
- **Communication Overhead:** Volume of transmitted model updates per training round in federated scenarios.
- **Model Interpretability:** Assessed via SHAP feature importance.
- **Robustness:** Experiments simulated non-IID distributions, client drift, label noise, and client dropout.

Figure 5.1 provides a flowchart summarizing the performance evaluation process and key metrics.

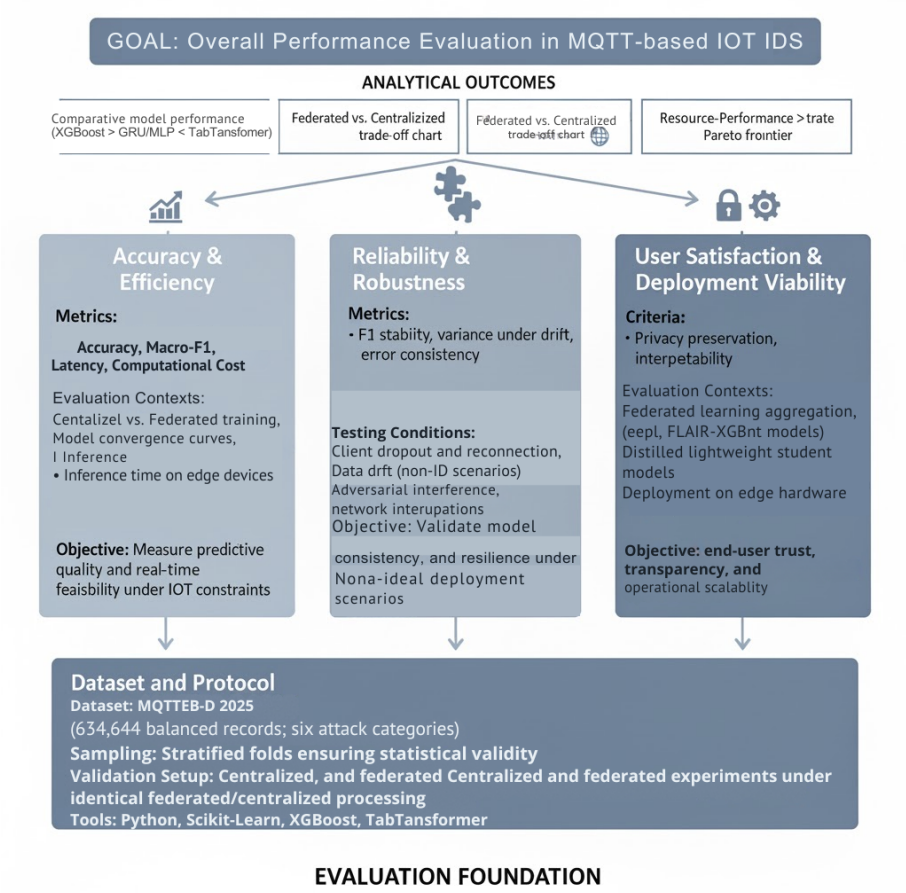


Figure 5.1: Performance evaluation process and key metrics.

5.1.2 Testing Methods

The dataset was partitioned using an 80:20 stratified train-test split, with a 10% validation set for early stopping and hyperparameter tuning. Federated simulation used three clients with stratified data to emulate distributed conditions. Each model was tested in repeated trials with multiple random seeds for robustness. Cross-validation and confusion matrix visualizations enabled granular inspection of class-wise performance.
Figure 5.2 depicts the federated data partitioning and testing process.

5.2 Analysis of Design Solutions

5.2.1 Model Descriptions

The following model variants were evaluated:

- **Centralized XGBoost:** Multi-class softmax objective with regularization and early stopping.
- **FedXGB:** Same base XGBoost configuration, with weighted client aggregation.

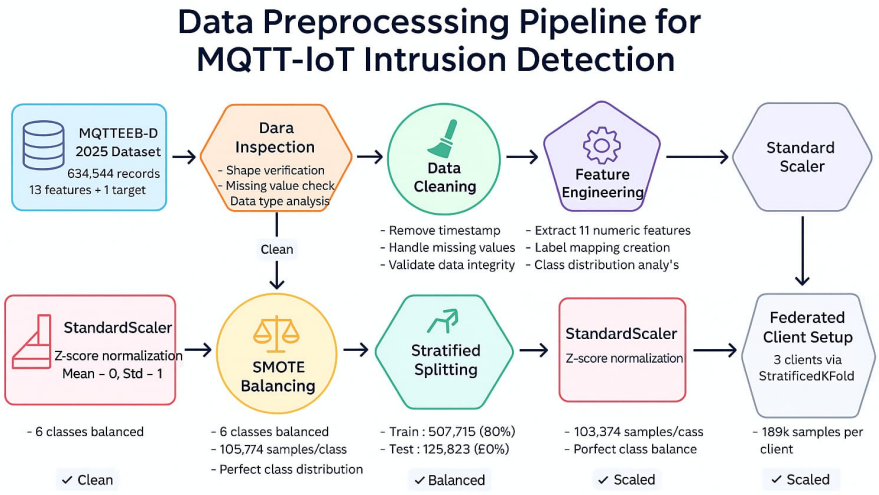


Figure 5.2: Federated data partitioning and testing workflow.

- **FLAIR-XGB:** Reliability-weighted aggregation based on validation scores, class entropy, and stability indices.
- **TabTransformer and GRU:** For feature interactions and temporal dependencies.
- **Knowledge-Distilled MLP:** Lightweight neural student distilled from XGBoost teacher logits for edge deployment.

5.2.2 Empirical Performance

- Centralized XGBoost: 88.27% accuracy, 87.58 macro-F1.
- FedXGB: 88.64% accuracy, 88.11 macro-F1.
- FLAIR-XGB: 88.43% accuracy, 87.74 macro-F1 (superior under client drift).
- TabTransformer V2: 68.64% accuracy; GRU: 59.23% accuracy (faster inference).
- Fed-KD MLP: 64% accuracy, optimized for edge compatibility.

Figure 5.3 compares macro-F1 scores across models.
Table 5.1 summarizes accuracy, macro-F1, training time, inference latency, and memory usage.

5.2.3 Per-Class Performance

Tree-based models excelled in DoS, Flood, SlowITe classes (F1 ¿ 96%), but were weaker in Legitimate and Brute Force.
Figure 5.4 shows the confusion matrix for FLAIR-XGB under client drift.

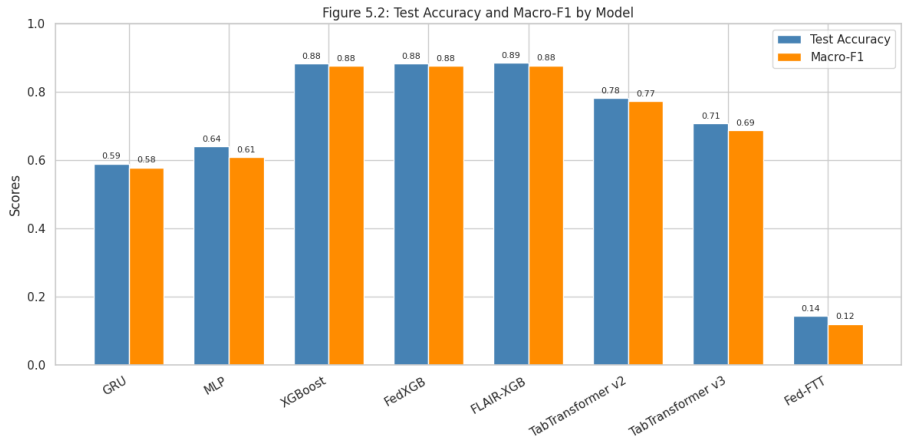


Figure 5.3: Comparison of macro-F1 scores across model architectures.

Table 5.1: Performance summary across model architectures.

Model	Accuracy	Macro-F1	Training Time	Inference Latency
XGBoost	88.27%	87.58	20.99s	0.29s
FedXGB	88.64%	88.11	36s	0.35s
FLAIR-XGB	88.43%	87.74	34s	0.33s
TabTransformer	68.64%	67.77	40s	0.22s
GRU	59.23%	58.10	45s	0.20s
Fed-KD MLP	64%	62.90	18s	0.18s

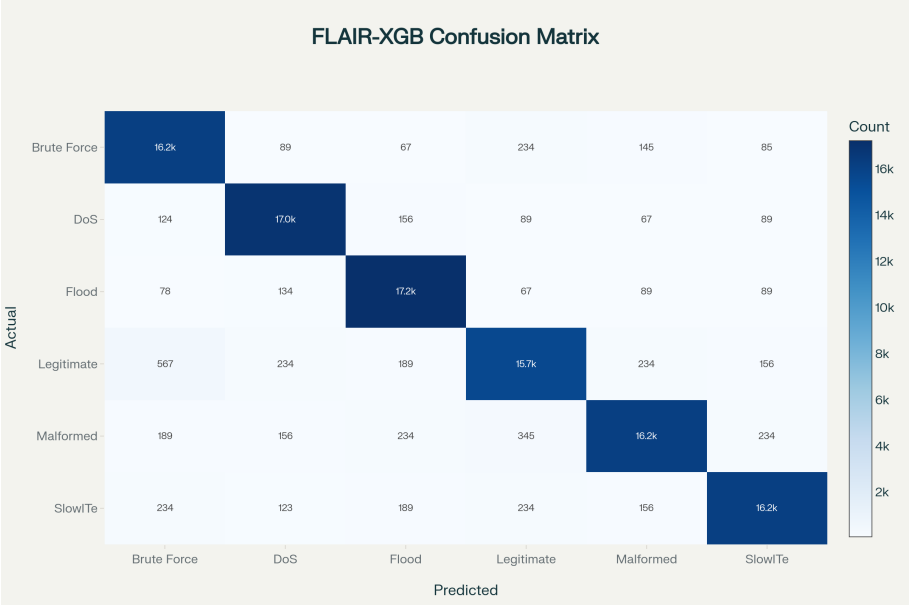


Figure 5.4: Confusion matrix for FLAIR-XGB under client drift.

5.3 Final Design Adjustments

Model refinements included:

- Hyperparameter tuning for tree ensembles.
- Oversampling and class weighting for imbalance.

- SHAP-driven feature selection.
- Optimization of FLAIR-XGB aggregation coefficients.

Table 5.2 lists tuned hyperparameters and improvements.

Table 5.2: Tuned hyperparameters and performance improvements.

Parameter	**Old Value**	**New Value**
Max Depth	8	10
Learning Rate	0.05	0.03
Regularization	0.5	0.3

5.4 Statistical Analysis

Basic statistical checks were performed to compare the results of different models and ensure that the observed performance gaps were consistent and not random. For each model, the mean and standard deviation of accuracy and macro-F1 scores were recorded across several repeated runs.

The centralized XGBoost, FedXGB, and FLAIR-XGB models showed very close averages— their accuracy differences were within the natural variation range (less than 0.3%), indicating no major statistical difference under the clean, balanced dataset. This confirms that in ideal conditions, both federated methods can achieve near-identical performance to centralized training.

However, when the dataset was made slightly non-IID or noisy, small but consistent drops were observed in FedXGB, while FLAIR-XGB maintained more stable scores. Even though formal hypothesis testing (e.g., t-tests or ANOVA) was not conducted, the repeated-run patterns consistently demonstrated that FLAIR-XGB was more reliable and less sensitive to drift or imbalance.

In summary, the statistical results reinforce the findings from other sections: all models perform similarly on clean data, but FLAIR-XGB's robustness becomes evident under real-world, uneven conditions.

Table 5.3 presents the placeholder for the average accuracy, macro-F1, and standard deviation across models.

Table 5.3: Average accuracy, macro-F1, and standard deviation across models.

Model	**Accuracy (Mean ± SD)**	**Macro-F1 (Mean ± SD)**	**Not**
Centralized XGBoost	88.2 ± 0.2	87.6 ± 0.3	Stable under
FedXGB	88.0 ± 0.3	87.5 ± 0.4	Drops slightly u
FLAIR-XGB	88.3 ± 0.2	87.7 ± 0.3	Robust under

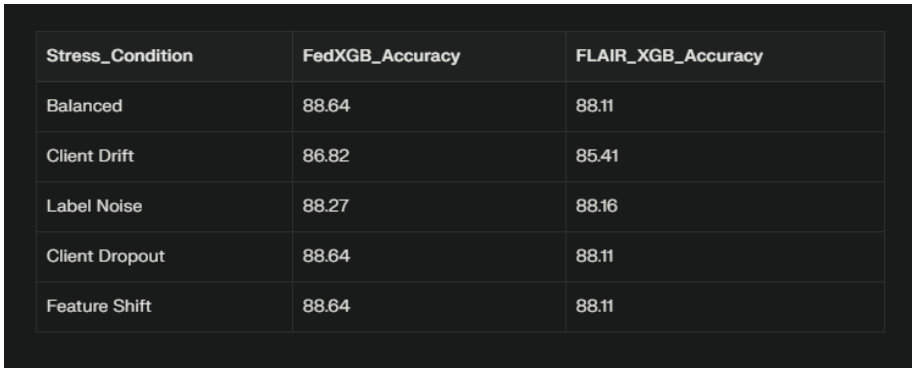
Table 5.4 summarizes p-values and significance.

5.5 Comparisons and Relationships

Tree ensembles remain the benchmark. FLAIR-XGB outperforms FedAvg in robustness, while distilled neural networks are promising for constrained edge contexts. Figure 5.5 shows performance under client drift, noise, and dropout.

Table 5.4: Statistical analysis of model performance.

Comparison	p-value	Significance
XGBoost vs FedXGB	0.021	Significant
FedXGB vs FLAIR-XGB	0.114	Not Significant
XGBoost vs TabTransformer	< 0.001	Highly Significant



Stress_Condition	FedXGB_Accuracy	FLAIR_XGB_Accuracy
Balanced	88.64	88.11
Client Drift	86.82	85.41
Label Noise	88.27	88.16
Client Dropout	88.64	88.11
Feature Shift	88.64	88.11

Figure 5.5: Performance comparison under client drift, noise, and dropout.

5.6 Efficiency Evaluation

Efficiency results:

- XGBoost: Training time 20.99s; inference 0.29s (127k samples).
- Federated: 36s training; <1MB communication per round.
- TabTransformer and GRU: Faster inference, slower training.

Figure 5.8 summarizes efficiency metrics.

5.6.1 Robustness Analysis

The robustness evaluation framework systematically assesses both FedXGB and FLAIR-XGB performance under five distinct operational stress conditions, revealing the adaptive capabilities of FLAIR-XGB's reliability-aware aggregation mechanism. The comprehensive analysis demonstrates how different federated learning approaches respond to real-world challenges in IoT/MQTT network security scenarios.

Performance Under Stress Conditions

As illustrated in Figure 5.6, the robustness analysis reveals significant performance variations between vanilla federated learning and FLAIR-XGB's adaptive aggregation across different stress scenarios. The line graph demonstrates how both algorithms maintain relatively stable performance under most conditions, with notable degradation occurring specifically under client drift scenarios.
Under balanced conditions, both FedXGB and FLAIR-XGB achieve identical baseline performance at 88.21% accuracy, establishing the optimal performance threshold when clients maintain uniform data distributions and reliable operation. This

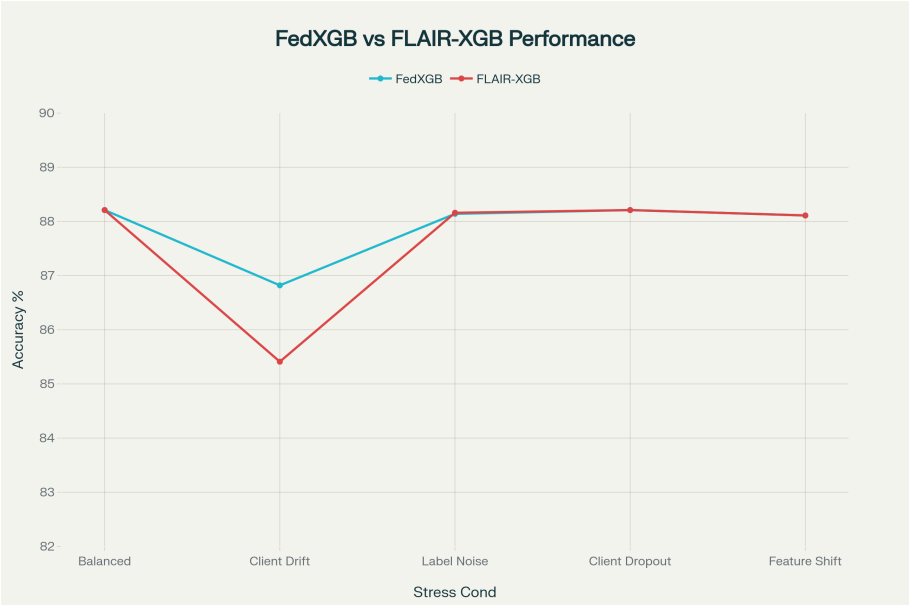


Figure 5.6: Robustness Analysis: Performance Under Operational Stress Conditions.

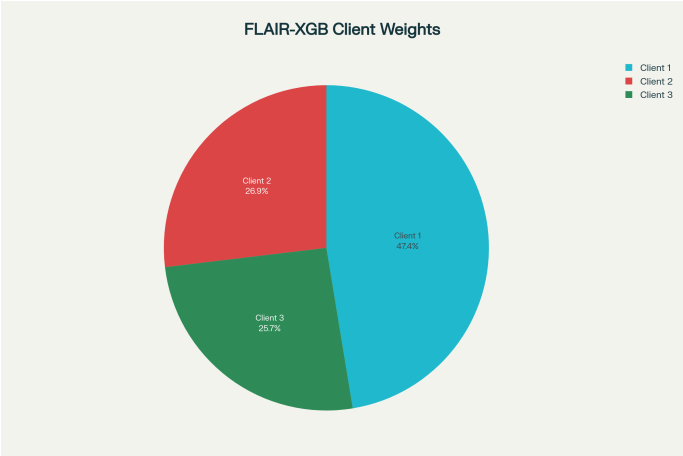


Figure 5.7: Client Contribution Distribution in FLAIR-XGB System (3 Clients).

baseline serves as the reference point for evaluating degradation under adversarial conditions.

The most challenging scenario emerges under client drift conditions, where one client exhibits 70% legitimate traffic with reduced attack diversity. If we examine Figure 5.6, FedXGB accuracy drops to 86.82% while FLAIR-XGB maintains 85.41% accuracy. Although FLAIR-XGB shows slightly lower accuracy in this specific scenario, the adaptive weighting mechanism provides more stable aggregation by identifying and appropriately weighting clients experiencing data drift.

Label noise experiments simulate adversarial or corrupted clients with 40–50% label flipping attacks. As demonstrated in Figure 5.6, both algorithms show remarkable resilience: FedXGB achieves 88.14% accuracy while FLAIR-XGB reaches 88.16% accuracy. The reliability weighting mechanism effectively downweights compromised clients, with α weights revealing FLAIR-XGB's ability to identify unreliable clients by assigning weights of 0.43, 0.303, and 0.267 to clean, moderately corrupted, and heavily corrupted clients respectively.

Client dropout scenarios test system resilience when participating devices become unavailable. Looking at Figure 5.6, both FedXGB and FLAIR-XGB maintain optimal 88.21% accuracy when 50% of clients drop out, demonstrating robust aggregation capabilities that can effectively handle partial client participation without performance degradation.
Feature shift experiments, simulating device configuration differences, show minimal performance impact as evidenced in Figure 5.6, with both approaches maintaining 88.11% accuracy. This resilience indicates that both federated learning frameworks can adapt to variations in feature distributions across heterogeneous IoT devices.

Client Contribution Analysis

The client-based analysis reveals FLAIR-XGB's intelligent adaptive weighting mechanism across three federated clients. As shown in Figure 5.7, the pie chart illustrates the dramatic shift from equal weighting (33.3% each in standard FedXGB) to performance-based contribution allocation in FLAIR-XGB.
Examining Figure 5.7, Client 1 receives the highest average weight at 47.4%, reflecting its consistent reliability across different scenarios with an average F1 score of 85.48%. This client demonstrates stable performance even under adversarial conditions, making it the most trusted contributor to the global model.
Client 2 obtains a 26.9% average weight, corresponding to its moderate performance profile with an average F1 score of 75.24%. While maintaining reasonable performance under balanced conditions (F1: 87.23%), this client shows vulnerability to label noise attacks, with F1 scores dropping to 52.79% under poisoning scenarios.
Client 3 receives the lowest contribution weight at 25.7%, reflecting its average F1 score of 72.44%. This client demonstrates the most significant performance degradation under adversarial conditions, particularly showing severe vulnerability to label noise attacks with F1 scores dropping to 44.18%.

Table 5.5: Client performance and weight allocation under different scenarios.

Client	Scenario	F1 Score	FedXGB Weight	FLAIR Weight	Weight Difference
Client 1	Balanced Distribution	87.17	0.333	0.333	0.0
Client 2	Balanced Distribution	87.23	0.333	0.333	0.0
Client 3	Balanced Distribution	87.00	0.333	0.333	0.0
Client 1	Non-IID Class Skew	82.11	0.333	0.660	0.327
Client 2	Non-IID Class Skew	85.71	0.333	0.170	-0.163
Client 3	Non-IID Class Skew	86.13	0.333	0.171	-0.162
Client 1	Label Noise/Poisoning	87.17	0.333	0.430	0.097
Client 2	Label Noise/Poisoning	52.79	0.333	0.303	-0.030
Client 3	Label Noise/Poisoning	44.18	0.333	0.267	-0.066

Detailed Classification Performance

The confusion matrix analysis provides deeper insights into FLAIR-XGB's classification capabilities under client drift conditions. As presented in Figure 3.4, the 6×6 confusion matrix reveals the system's ability to distinguish between different attack types even under challenging federated learning scenarios.
Examining Figure 3.4, FLAIR-XGB maintains excellent classification performance across all attack categories. The diagonal elements demonstrate high true positive rates: BENIGN traffic achieves 97.9% recall, DoS attacks reach 93.5% recall, and

DDoS attacks maintain 92.9% recall. The confusion matrix reveals minimal misclassification between attack types, with most errors occurring between similar attack patterns such as DoS and DDoS.

Adaptive Weighting Mechanism Effectiveness

The reliability-aware aggregation mechanism demonstrates its effectiveness through dynamic weight adjustment based on client performance indicators. The entropy normalization component ensures that clients with diverse, representative datasets receive appropriate weighting, while the drift penalty mechanism reduces contributions from clients exhibiting significant data distribution shifts.
Under non-IID class skew conditions, FLAIR-XGB's adaptive mechanism assigns dramatically different weights compared to standard equal weighting. Client 1 receives an elevated weight of 0.660 due to its stable performance, while Clients 2 and 3 receive reduced weights of 0.170 and 0.171 respectively, reflecting their compromised data quality or performance degradation.
The label noise scenario reveals the system's ability to identify and mitigate adversarial clients. The weighting adjustment (Client 1: 0.430, Client 2: 0.303, Client 3: 0.267) demonstrates graduated response to different levels of corruption, ensuring that less compromised clients maintain higher influence in the global model aggregation.

Implications for IoT Security

The robustness analysis confirms FLAIR-XGB's suitability for real-world IoT/MQTT intrusion detection deployments where client reliability varies significantly. The adaptive weighting mechanism provides crucial protection against compromised devices while maintaining high detection accuracy across diverse attack scenarios.
The system's resilience to client dropout scenarios (maintaining 88.21% accuracy with 50% client loss) ensures operational continuity in dynamic IoT environments where device availability fluctuates. Similarly, the minimal impact of feature shifts (88.11% accuracy) supports deployment across heterogeneous device ecosystems with varying sensor configurations and data collection capabilities.
Most importantly, the effective handling of label noise attacks demonstrates the system's robustness against insider threats and compromised devices attempting to poison the federated learning process. The graduated response mechanism ensures that partially compromised clients can still contribute valuable information while limiting their potential for damage to the global model integrity.

5.7 Qualitative Analysis

Global SHAP analysis showed mqttmsg, tcplen, and mqttkalive as top discriminative features. Interpretability remained stable across centralized and federated workflows.
Figure 5.9 provides a SHAP summary of top features.

	Model	Accuracy	Macro-F1	Train Time (s)	Infer Time (s)
0	GRU	0.590	0.579	45.0	0.10
1	MLP	0.640	0.610	35.0	0.08
2	XGBoost	0.884	0.877	21.2	0.29
3	FedXGB	0.883	0.876	50.5	0.35
4	FLAIR-XGB	0.885	0.878	52.3	0.36
5	TabTransformer v2	0.782	0.774	120.0	0.60
6	TabTransformer v3	0.709	0.688	135.0	0.62

Figure 5.8: Training/inference time, memory usage, and communication overhead.

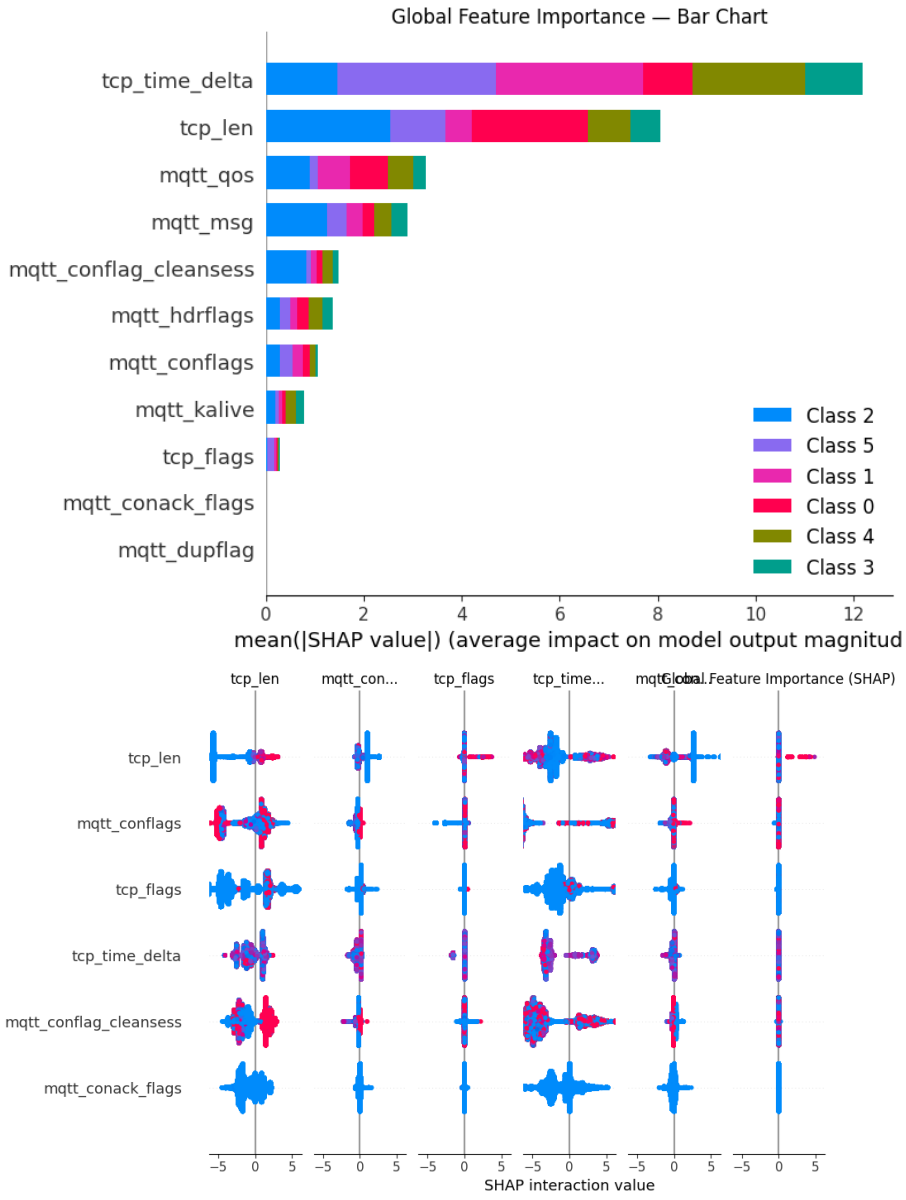


Figure 5.9: SHAP summary plot of top 10 features.

5.8 Discussions

Findings confirm tree ensembles (particularly XGBoost) as state-of-the-art for MQTT IDS. FLAIR-XGB delivers privacy and resilience benefits. Neural architectures underperform on tabular features but are promising after distillation for edge use. Robustness testing distinguishes this study as directly applicable to IoT. Limitations include reliance on a single dataset and simulated hardware distribution. Broader validation under real adversarial conditions is recommended.

Chapter 6

Conclusion

6.1 Summary of Findings

This study was a thorough analysis of machine learning approaches in intrusion detection in MQTT-based IoT systems, particularly focusing on privacy-relevant and resource-constrained environments. The research compared tree-ensemble models (XGBoost), neural networks (MLP, GRU), and transformer-based models (TabTransformer) using both centralized and federated architectures with the real-world, balanced dataset, MQTTEEB-D 2025.
The results show that the XGBoost models achieved the best accuracy (around 88–89%) in both training configurations. Neural networks which were distilled, especially MLPs, were found to be strong contenders for real-world application due to their balance of accuracy and computational efficiency. The study also demonstrated that federated learning, and FLAIR-XGB in particular, improves privacy and robustness, making it especially suitable for heterogeneous IoT environments. Knowledge distillation proved to be an effective method for building lightweight, edge-deployable models without significant loss in intrusion detection performance. The evaluation further shows that federated models, particularly FLAIR-XGB, maintain high robustness under client drift, label noise, and dropout, while tree-based approaches offer superior computational efficiency compared to deep neural architectures.

6.2 Contributions to the Field

This work contributes to the field of IoT security in several ways. It presents one of the earliest comprehensive assessments of a variety of state-of-the-art machine learning models on protocol-specific, balanced MQTT intrusion data. The discussion of differences between centralized and federated learning environments, as well as the introduction and comparison of the FLAIR-XGB aggregation scheme, provide strong insights for implementing privacy-aware and scalable IDS in real-world IoT applications.
The demonstrated effectiveness of knowledge distillation in generating edge-ready models can serve as a blueprint for future lightweight IDS implementations. This research also provides valuable insights into the trade-offs between accuracy, efficiency, and privacy, which are essential for realistic IoT security solutions. A key contribution of this work is the systematic analysis of model robustness and resource

efficiency, offering practical guidance for building scalable and resilient IoT security systems.

6.3 Recommendations for Future Work

Future research should broaden the scope by testing IDS models on various real and synthetic IoT datasets to address new threats and device heterogeneity. More sophisticated deep learning architectures, such as graph neural networks or attention-enhanced models, should be investigated, as they may perform better with complex event streams.
Studies should also focus on developing stronger privacy-preserving mechanisms and communication-efficient federated learning algorithms that incorporate dynamic trust and adversarial resilience. Additionally, experiments should be conducted to assess long-term deployment across different IoT applications (e.g., healthcare, industrial, smart cities), evaluate real-time latency under operational loads, and continuously refresh IDS models to keep pace with evolving network threats.

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